College of Engineering Trivandrum

Department of Computer Applications

20MCA203 . Design and Analysis of Algorithms (Semester 3)

Assignment I

1. What is an Approximation algorithm? Explain the approximation ratio for minimization and maximization problems with suitable examples.

2. Describe the 2-approximation algorithm for the Vertex Cover problem. Prove that the solution obtained is at most twice the size of an optimal vertex cover.

3. Formulate the Vertex Cover problem as an Integer Linear Programming problem.

4. Explain LP-rounding algorithm for the Vertex Cover problem.